CHRIS BASTING

EDUCATION:

Ph.D. – Bioinformatics and Computational Biology **Minneapolis, MN**
University of Minnesota Twin Cities *August 2026*
Dissertation Title: Investigating Chronic Inflammation in People with HIV on Antiretroviral Therapy
Advisor: Dr. Nichole Klatt, Ph.D.

M.S. – Bioinformatics and Computational Biology **Minneapolis, MN**
University of Minnesota Twin Cities *August 2023*

B.S. – Biochemistry **Minneapolis, MN**
University of Minnesota Twin Cities *May 2015*

PROFESSIONAL EXPERIENCE:

Researcher Manager 2 **Minneapolis, MN**
Laboratory of Dr. Nichole Klatt, University of Minnesota *July 2020 – Present*

- Oversee daily lab operations, personnel, budget, and compliance with IRB/IBC and SOP standards. Recruit, mentor, and evaluate staff.
- Design and conduct molecular/microbiological experiments (DNA/RNA extraction, ELISAs, bacterial culture, LC-MS/MS) and develop bioinformatics pipelines for multi-omic data integration and visualization.
- Lead grant and manuscript authorship, present findings at scientific meetings and cross-institution collaborations.
- Contributed to numerous studies and publications investigating host-microbe interactions in recurrent bacterial vaginosis and infectious diseases such as HIV, COVID-19, and Zika.

Service Coordinator **Minneapolis, MN**
Diversigen *May 2019 – June 2020*

- Managed next-generation sequencing microbiome projects end-to-end, coordinating Sales, Operations, and clients to align timelines and deliverables.
- Oversaw sample intake, LIMS documentation, and data integrity across all project stages.
- Provided scientific/technical support and delivered final sequencing result reviews to clients.

Research Scientist **Minneapolis, MN**
Diversigen *March 2018 – May 2019*

- Processed client samples through NGS workflows per SOPs, ensuring accurate tracking via Diversigen LIMS.
- Troubleshot instrumentation and optimized workflows for low-biomass samples; led process improvements including custom LIMS development.
- Authored SOPs and trained new lab staff.

Fermentation Lab Technician **Minneapolis, MN**
NatureWorks *February 2017 – March 2018*

- Developed and optimized fermentation processes converting methane to lactic acid, running weekly trials across strains/conditions to maximize yield.

- Designed PCR primers and used Sanger sequencing to characterize genes linked to lactic acid tolerance.
- Analyzed fermentation broths via HPLC/GC and operated continuous culture systems (chemostat/turbidostat) for adaptive evolution studies.

Teaching Assistant **Minneapolis, MN**
University of Minnesota *January 2014 – May 2015*

- Held office hours, graded assignments/exams, and proctored exams for Biochemistry 3021

Laboratory Assistant **Minneapolis, MN**
University of Minnesota *September 2013 – June 2014*

- Prepared lab supplies, reagents, and equipment for student coursework for Biochemistry labs 4025 & 4125

Research Assistant **Minneapolis, MN**
University of Minnesota *June 2013 – August 2013*

- Studied adaptive evolution in *Vibrio fischeri* under Dr. William Soto, performing plating, colony counting, and media preparation.

AWARDS AND HONORS:

New Investigator Scholarship. Conference on Retroviruses and Opportunistic Infections 2026
Honorable mention, Best Oral Presentation, Department of Surgery Research Week 2025
University of Minnesota Genomics Center Pilot Sequencing Program 2024
New Investigator Scholarship. Conference on Retroviruses and Opportunistic Infections 2024
ICIS Student/Postdoc Member Travel Award. Cytokines Conference 2023
ICIS Student/Postdoc Member Travel Award. Cytokines Conference 2022

MENTORSHIP:

Graduate-level
2025-2026: Maura Barrett, master's student
2024-2026: Ty Schroeder, master's student

Undergraduate level
2025-present: Stella Dileva, student volunteer, Undergraduate Research Opportunity (UROP) mentor
2025-present: Annaka Binner, student volunteer
2024-present: Josh Knapp, student volunteer, Undergraduate Research Opportunity (UROP) mentor
2024-2026: Elaina Creagan, student volunteer
2025: Alex Litterer, student volunteer
2024-2025: Alex Rapp, student volunteer, Undergraduate Research Opportunity (UROP) mentor
2023-2025: Sophie Lebakken, Undergraduate Research Opportunity (UROP) mentor
2020-2022: Nicholas Dahl, student volunteer
2020-2022: Caroline Pinkerman, student volunteer

PUBLICATIONS: Refereed Journal Articles

1. **C. M. Basting**, R. Langat, C. A. Broedlow, C. R. Guerrero, T. D. Bold, M. Bailey, A. Velez, T. Schroeder, J. Short-Miller, R. Cromarty, Z. J. Mayer, P. J. Southern, T. W. Schacker, S. E. Safo, C. T. Bramante, C. J. Tignanelli, L. Schifanella, N. R. Klatt, SARS-CoV-2 infection is associated with intestinal permeability, systemic inflammation, and microbial dysbiosis in hospitalized patients. Microbiol. Spectr. 0, e00680-24 (2024).

2. **C. M. Basting**, J. Anderson, K. Escandón, G. Wieking, C. Guerrero, J. Reichel, R. T. Cromarty, E. Swanson, T. Schroeder, E. Creagan, M. Barrett, F. Torres-Ruiz, M. Soto-Nava, Lady Carvajal-Ruiz, K. K. Ordaz-Candelario, O. Briceño, N. Funderburg, M. Graham, P. Hunt, S. Avila-Rios, G. S. M. de Oca, T. W. Schacker, N. R. Klatt, Multi-omics links microbial dysbiosis, systemic inflammation, and metabolomic disruptions to SNAE risk in treated HIV. JCI Insight 11 (2026), doi:10.1172/jci.insight.205379.

3. **C. M. Basting**, T. A. Schroeder, K. G. Ferbas, R. R. Shields-Cutler, N. H. Tobin, A. Chakrawarti, A. Velez, E. Swanson, C. A. Broedlow, R. Langat, R. Cromarty, L. Schifanella, C. T. Bramante, G. M. Aldrovandi, A. Rimoin, O. O. Yang, J. A. Fulcher, N. R. Klatt, Gut barrier integrity biomarkers are associated with increased inflammation and predict disease status in hospitalized COVID-19 patients. Sci. Rep. (2026), doi:10.1038/s41598-026-59934-5.

4. **C. M. Basting**, T. A. Schroeder, R. Shields-Cutler, E. Swanson, C. Guerrero, C. R. Hemmila, C. A. Broedlow, A. Chakrawarti, A. Velez, R. Cromarty, A. Riva, A. Torre, A. Lai, L. Schifanella, N. R. Klatt, Longitudinal multi-omic dynamics in hospitalized COVID-19 patients based on disease severity. Gut Microbes Rep. 3, 2712728 (2026).

5. **C. M. Basting**, N. R. Klatt, Dissecting the Impact of the Gut Microbiome on HIV Reservoir Dynamics. J. Infect. Dis. 233, 622–624 (2026).

6. E. C. Swanson, **C. M. Basting**, N. R. Klatt, The role of pharmacomicrobiomics in HIV prevention, treatment, and women's health. Microbiome 12, 254 (2024).

7. J. Rhein, J. G. Chipman, G. J. Beilman, R. Cromarty, K. Escandón, J. Anderson, G. Wieking, J. Reichel, R. Batres, A. Khoruts, **C. M. Basting**, P. Hinderlie, Z. B. Davis, A. Eaton, B. P. Vaughn, E. Eilkhani, J. T. Safrit, P. Soon-Shiong, J. V. Baker, N. R. Klatt, S. G. Deeks, J. S. Miller, T. W. Schacker, Impact of the IL-15 superagonist N-803 on lymphatic reservoirs of HIV. JCI Insight 10 (2025), doi:10.1172/jci.insight.190831.

8. C. Martino, B. P. Kellman, D. R. Sandoval, T. M. Clausen, R. Cooper, A. Benjdia, F. Soualmia, A. E. Clark, A. F. Garretson, C. A. Marotz, S. J. Song, S. Wandro, L. S. Zaramela, R. A. Salido, Q. Zhu, E. Armingol, Y. Vázquez-Baeza, D. McDonald, J. T. Sorrentino, B. Taylor, P. Belda-Ferre, P. Das, F. Ali, C. Liang, Y. Zhang, L. Schifanella, A. Covizzi, A. Lai, A. Riva, **C. Basting**, C. A. Broedlow, A. S. Havulinna, P. Jousilahti, M. Estaki, T. Kosciolek, R. Kuplicki, T. A. Victor, M. P. Paulus, K. E. Savage, J. L. Benbow, E. S. Spielfogel, C. A. M. Anderson, M. E. Martinez, J. V. Lacey, S. Huang, N. Haiminen, L. Parida, H.-C. Kim, J. A. Gilbert, D. A. Sweeney, S. M. Allard, A. D. Swafford, S. Cheng, M. Inouye, T. Niiranen, M. Jain, V. Salomaa, K. Zengler, N. R. Klatt, J. Hasty, O. Berteau, A.

F. Carlin, J. D. Esko, N. E. Lewis, R. Knight, SARS-CoV-2 infectivity can be modulated through bacterial grooming of the glycocalyx. mBio 0, e04015-24 (2025).

9. T. A. Schroeder, J. A. Manuzak, C. J. Miller, A. T. Gustin, **C. M. Basting**, R. K. Cheu, A. Velez, C. B. Driscoll, J. Tisoncik-Go, L. Schifanella, T. Hensley-McBain, C. A. Evandy, E. A. Smith, D. Bratt, R. Shields-Cutler, J. Smedley, D. H. Fuller, D. H. Barouch, M. A. O'Connor, M. Gale, N. R. Klatt, Elevated peripheral and nervous system inflammation is associated with decreased short-chain fatty acid levels in Zika virus-infected macaques. J. Virol. 99, e0100325 (2025).

10. C. A. Broedlow, E. Swanson, E. M. Cherenack, **C. Basting**, N. F. Nogueira, M. France, P. Yue, A. Chakrawarti, A. Salazar, L. Acosta, P. Raccamarich, M. Gale, J. Ravel, L. A. Fein, J. Holm, M. L. Alcaide, N. R. Klatt, Common cervicovaginal sequencing methods result in discordant molecular diagnoses of bacterial vaginosis and reveal strain level effects of Gardnerella vaginalis. Npj Womens Health 3, 41 (2025).

11. A. Chakrawarti, R. T. Cromarty, **C. M. Basting**, J. Anderson, T. A. Schroeder, K. Escandón, R. Shields-Cutler, R. Langat, E. Swanson, P. Soon-Shiong, J. T. Safrit, L. S. Sender, S. Reddy, J. S. Miller, J. Rhein, T. W. Schacker, N. R. Klatt, Pre-treatment Gut Microbiome Diversity and Function Linked to Cytotoxic and Natural Killer Cell Immune Responses after N-803 Treatment in People with HIV. Clin. Infect. Dis. , ciag369 (2026).

OTHER PUBLICATIONS: Preprints and Manuscripts in Progress or Under Review

1. **C. M. Basting**, C. Guerrero, K. Escandón, J. Anderson, G. Wieking, E. Swanson, T. Schroeder, C. Hemmila, R. T. Cromarty, F. Torres-Ruiz, M. Soto-Nava, L. Carvajal-Ruiz, K. Ordaz-Candelario, O. Briceño, N. Funderburg, S. Avila-Rios, M. L. Graham, T. W. Schacker, G. Salgado Montes de Oca, N. R. Klatt, Plasma metabolomics reveals lipid-predominant metabolic disruptions in virologically suppressed people with HIV. *Under Review at Cell Reports Medicine*

2. N. Klatt, E. Swanson, C. Broedlow, E. Cherenack, N. Nogueira, **C. Basting**, P. Yue, A. Chakrawarti, T. Schroeder, A. Salazar, L. Acosta, P. Raccamarich, M. Gale, L. Fein, M. Alcaide, Host-microbe interactions characterized by gene expression of cervical adhesion molecules, cytokines, and growth factors define the recurrence of bacterial vaginosis. Res. Sq. , rs.3.rs-7576831 (2025). *Under review at Nature Communications*.

3. T. A. Schroeder, **C. M. Basting**, E. Swanson, M. Bailey, A. Velez, J. Li, C. Broedlow, T. W. Schacker, M. Osborn, N. R. Klatt, Unique microbiome signatures in wastewater are predictive of COVID-19 case rates. *Under review at Applied and Environmental Microbiology*.

4. P. A. Crawford, J. Rhein, J. G. Chipman, G. J. Beilman, R. Cromarty, K. Escandón, J. Anderson, G. Wieking, A. Johnston, A. Ahmed, J. Reichel, A. Khoruts, **C. M. Basting**, N. Kuchma, J. V. Baker, N. R. Klatt, A. T. Haase, T. W. Schacker, Ongoing lymphoid HIV production drives pyroptosis and GLP-1 counter-regulation in ART-suppressed infection. BioRxiv Prepr. Serv. Biol. , 2026.01.09.698696 (2026).

5. M.K. Rothenberger, J. Anderson, K. Escandon, **C.M. Basting**, G. Wieking, J. Reichel, A. Thorkelson, S. Jorstad, C. David, G.J. Beilman, J.G. Chipman, T Hoskuldsson, J. Sanchez, N.R.

Klatt, J.V. Baker, J.P. Casazza, R.A. Koup, D.C. Douek, C.V. Fletcher, T.W. Schacker, Losartan to improve immune reconstitution in HIV disease. *Under Review at JCI Insight*.

6. A. Chakrawarti, E. Swanson, C.A. Broedlow, Y. Pan, E.M. Cherenack, N.F. Nogueira, **C.M. Basting**, L. Acosta, P. Raccamarich, L.A. Fein, M.L. Alcaide, N.R. Klatt, Distinct Antimicrobial Resistance Trajectories Distinguish Persistent from Recurrent Bacterial Vaginosis. *In progress*.

7. M Barrett, J. Anderson, K. Escandón, G. Weiking, T. Schroeder, E. Swanson, M. L. Graham, M. Gale, T. W. Schacker, N. R. Klatt, **C. M. Basting**, Spatial Transcriptomics Reveals Compartment-Specific Immune Activation Signatures in Ileal and Lymph Node Tissue in Treated HIV Infection. *In progress*.

8. C. A. Broedlow, E. Swanson, E. M. Cherenack, N. Fonseca Nogueira, **C. M. Basting**, A. Salazar, L. Acosta, P. Raccamarich, L. Fein, J. B. Holm, M. L. Alcaide, N. R. Klatt, Temporal Dynamics of Vaginal Microbiota in Amsel-defined Recurrent Bacterial Vaginosis. *In progress*.

ABSTRACTS:

Oral Presentations:

C. M. Basting, K. Escandon, J. Anderson, R. Cromarty, T. Schroeder, A. Velez, G. Wieking, J. Reichel, C. Guerrero, O. Briceño, F. Torres-Ruiz, S. Avila, G. Salgado, T. Schacker, N. R. Klatt. Specific gut microbes are associated with immune activation in people living with HIV. *Cytokines Conference. Athens, Greece, (2023, October 15–18)*.

C. M. Basting. Investigating the stool and plasma metabolome of people with HIV. *University of Minnesota Center for Metabolomics and Proteomics, Lunch and Learn, (2024, October 10th)*. Invited Speaker.

C. M. Basting. Investigating mechanisms of chronic inflammation in people with HIV. *Biocrates breakfast seminar, American Society of Mass Spectrometry, (2025, June 3rd)*. Invited Speaker.

C. M. Basting. Investigating mechanisms of chronic inflammation in people with HIV. *University of Minnesota, Department of Surgery Research Week, (2025, May 18th)*.

C. M. Basting. Targeted Metabolomics Reveals the Metabolic Impact of HIV Infection Across Viral Control States. *University of Minnesota Center for Metabolomics and Proteomics, MassSpectacular, (2026, April 14th)*. Invited Speaker.

Poster Presentations:

C. M. Basting, R. Cromarty, J. Short-Miller, M. Ahmad, C. Broedlow, R. Cheu, B. Witthuhn, C. Guerrero, T. Griffin, L. Schifanella, N. R. Klatt. Identifying bacteria-drug interactions between HIV antiretrovirals and the gut microbiome. *American Society for Mass Spectrometry. Convention Center, Minneapolis, MN, (2022, June 5–9)*.

C. M. Basting, C. Broedlow, R. Cromarty, J. Short-Miller, A. Velez, J. Ling, L. Schifanella, N. R. Klatt. Inflammatory cytokines are correlated with microbial dysbiosis and disease severity in hospitalized COVID-19 patients. *Cytokines Conference. Hilton Hotel, Big Island, HI, (2022, September 20–23)*.

C. M. Basting, C. Broedlow, R. Cromarty, J. Short-Miller, A. Velez, J. Ling, L. Schifanella, N. R. Klatt. Inflammatory cytokines are correlated with microbial dysbiosis and disease severity in hospitalized COVID-19 patients. *University of Minnesota, Department of Surgery Research Week, (2022, October 4th)*.

T. Schroeder, **C. M. Basting**, K. Ferbas, A. Velez, C. Broedlow, E. Swanson, M. Bailey, R. Langat, L. Schifanella, N. Tobin, G. Aldrovandi, O. Yang, J. Fulcher, N. R. Klatt. Markers of microbial translocation and inflammatory cytokines are predictive of severe COVID-19. *Conference on Retroviruses and Opportunistic Infections. Colorado Convention Center, Denver, CO, (2024, March 3–6)*.

C. M. Basting, T. Schroeder, A. Velez, C. Broedlow, P. Hunt, A. Collier, T. Griffin, C. Guerrero, N. R. Klatt. Microbe-Drug Interactions Between Antiretrovirals and the Gut Microbiome in HIV Infections. *Conference on Retroviruses and Opportunistic Infections. Colorado Convention Center, Denver, CO, (2024, March 3–6)*.

S. Lebakken, **C. M. Basting**, M. Bailey, T. Schroeder, C. Broedlow, C. Guerrero, C. Hemmila, N. R. Klatt. Investigating the Effects of Antidepressants on Intestinal Bacteria. *National Conference on Undergraduate Research. Long Beach Convention Center, Long Beach, CA, (2024, April 8–10)*.

C. Hemmila, **C. M. Basting**, S. Lebakken, M. Bailey, E. Swanson, C. Broedlow, T. Schroeder, T. Griffin, E. Cherenack, N. R. Klatt, C. Guerrero. Antidepressant screening via Andrew+ sample preparation automation and LC-MS/MS to access microbiota and drug interactions. *American Society for Mass Spectrometry. Anaheim Convention Center, Anaheim, CA, (2024, June 3–6)*.

T. Schroeder, **C. Basting**, M. Bailey, T. Schacker, M. Osborn, N. Klatt. Microbiome signature in wastewater is predictive of COVID-19 case rate. *Microbiome and Cancer Symposium, (2024, September 13th)*.

T. A. Schroeder, **C. M. Basting**, E. Swanson, A. Chakrawarti, L. Grafstrom, A. Organ, M. M. Anderson, A. B. Tuma, M. Pritzker, H. Lim, F. Ikramuddin, D. Zachs, N. R. Klatt. Associations between the gut microbiome and Long COVID patient response to ultrasound treatment. *Microbiome and Cancer Symposium, (2025, October 17th)*.

C. M. Basting, J. Anderson, K. Escandon, G. Wieking, J. Reichel, R. Cromarty, T. Schroeder, E. Swanson, A. Chakrawandi, S. Avila-Rios, N. Funderburg, P. Hunt, G. Salgado, T. W. Schacker, N. R. Klatt. Multi-omics Reveals Metabolic and Microbial Markers of Serious Non-AIDS Event Risk in People with HIV. *Conference on Retroviruses and Opportunistic Infections. Colorado Convention Center, Denver, CO, (2026, February 22–25)*.

C. R. Hemmila, **C. M. Basting**, T. J. Griffin, N. R. Klatt, C. R. Guerrero. Antiretroviral screening via LC-MS/MS to assess microbiota and drug interactions. *American Society for Mass Spectrometry, San Diego Convention Center, CA, (2026, May 31–June 4)*.